

A Square-Zero Counterexample to Conjecture 2.1 in *A Conditional Proof of the ISP for Quasinilpotent Operators*

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August 11, 2026

Status. Candidate counterexample, likely valid, pending expert review. Conjecture 2.1 of Norman's arXiv:2008.03253v2 is false as printed. The construction is a rank-one square-zero operator supported on a two-dimensional subspace of the required infinite-dimensional Hilbert space.

1 Source conjecture

Norman [1] works on a complex separable infinite-dimensional Hilbert space. His Conjecture 2.1 begins with a quasinilpotent operator T and asserts the existence of $a, b, t > 0$. Write $\tilde{T} = mT$, where $m > 0$ is chosen so that $\|\tilde{T}\| = a$. For every rank-one quasinilpotent F of norm b such that $\tilde{T}^* + \alpha F$ is nonquasinilpotent for at least two distinct complex values of α , the conjecture requires a set $S \subset \mathbb{C} \setminus \{0\}$, $|S| \geq 2$, such that:

(i) $\tilde{T}^* + \alpha F$ is nonquasinilpotent for all $\alpha \in S$;

(ii) for every $\alpha \in S$,

$$\sigma_t(\tilde{T} + \alpha F) \subseteq \sigma(\tilde{T} + \alpha F) + B(0, \Phi_{a,b,t,S}(\alpha)),$$

and the set on the right is disconnected in $\mathbb{C}$.

The counterexample below uses the fact that the hypothesis is imposed on the adjoint perturbation while the disconnectedness conclusion is imposed on the non-adjoint perturbation.

2 Proof intuition

The adjoint asymmetry can already be maximized by the smallest nilpotent shift. Let N send one basis vector to another and vanish everywhere else. Perturbing N by a scalar multiple of itself never changes square-zero nilpotence, so the non-adjoint spectrum in conclusion (ii) stays equal to $\{0\}$. In contrast, adding the same N to N^* closes a two-cycle. The resulting 2×2 block has two nonzero eigenvalues for every nonzero perturbation parameter. Thus the conjecture's hypothesis holds as strongly as possible, while the required thickened spectrum is always a disk and can never be disconnected.

3 Counterexample

Theorem 1. *Conjecture 2.1 of arXiv:2008.03253v2 is false as stated.*

Indeed, the proof of the upper semicontinuity of the spectrum given in [19] consists of three main steps [2](#): firstly, it is taken a value of $\delta > 0$ such that the curve Γ satisfies $\Gamma \subset P(B)$ (the resolvent set of B), and then Lemma 4.10 in Chapter I of [19] is applied (notice that the hypothesis that the projection depends *continuously* on the parameter is satisfied by Theorem 3.15). From this we get Theorem 3.16, which gives us Proposition [2.5](#) above via Theorem 2.14. Indeed, if $\|B\| < \delta$, then $\delta(A+B, A) \leq \|B\| < \delta$, so that Theorem 3.16 applies and leads to the desired result. Now, $\Gamma \subset P(B)$ certainly holds under some assumptions. In the proof given in [19], the general case is considered: given two *closed* operators A , B , and a curve Γ as above, by the proof of Theorem 3.1 we may take

$$\delta = \min_{\xi \in \Gamma} \frac{1}{2} \frac{1}{1 + |\xi|^2} \frac{1}{\sqrt{1 + \|R_A(\xi)\|^2}}$$

But in our case, we can be less general and get something easier to deal with. By Remark 3.2, if B is a *bounded* operator (as in our case), then we can take:

$$\delta = \min_{\xi \in \Gamma} \frac{1}{\|R_A(\xi)\|}$$

We conclude this section by stating the following conjecture, which is inspired by the results contained in [11] (in particular, by Theorem 6.1).

Conjecture 2.1. *Let T be quasinilpotent. Then, there exist $a, b, t > 0$ such that, for every rank 1 quasinilpotent operator F with $\|F\| = b$ for which $\tilde{T}^* + \alpha F$ is not quasinilpotent for at least two distinct complex α 's, there is some subset $S \subseteq \mathbb{C} \setminus \{0\}$ with $|S| \geq 2$ (which may depend on F) satisfying the following properties:*

- (i) $\tilde{T}^* + \alpha F$ is not quasinilpotent for all $\alpha \in S$;
- (ii) for all $\alpha \in S$, the following inclusion holds

$$\sigma_t(\tilde{T} + \alpha F) \subseteq \sigma(\tilde{T} + \alpha F) + B(0, \Phi_{a,b,t,S}(\alpha))$$

and $\sigma(\tilde{T} + \alpha F) + B(0, \Phi_{a,b,t,S}(\alpha))$ is disconnected in $\mathbb{C}$.

Here, $\tilde{T} = mT$, with $m > 0$ such that $\|\tilde{T}\| = a$, and Φ is a function depending on α .

Actually, it will be clear from the proof of our main result that even some slight modifications of this conjecture would imply the existence of a n.i.s. for every quasinilpotent operator.

Remark 2.2. We note that the first part of the conjecture is clearly true: the function Φ , which is not required to be continuous or to satisfy any other condition, always exists. Moreover, it is easy to construct examples of quasinilpotent operators A having some rank 1 perturbations $A + \alpha F$ for distinct values of α which are not quasinilpotent (use, e.g., Lemma 2.1 in [30]). So the actual

²If not explicitly stated otherwise, all the results to which we refer in this Remark are given in Chapter 4 of [19].

Figure 1: Conjecture 2.1 in Norman [1, p. 5]. The full readable page width is retained so that the adjoint in the hypothesis and its absence in conclusion (ii) can be checked directly.

Proof. Let $\mathcal{H}$ be a complex separable infinite-dimensional Hilbert space and fix orthonormal vectors e_1, e_2 . Define $N \in B(\mathcal{H})$ by

$$Ne_2 = e_1, \quad Ne_1 = 0, \quad N|_{\{e_1, e_2\}^\perp} = 0.$$

Then N has rank one, $N^2 = 0$, and $\|N\| = 1$. In particular, $T := N$ is quasinilpotent.

We show that no triple $a, b, t > 0$ can satisfy the conjecture for this T . Fix arbitrary $a, b, t > 0$. Since $\|T\| = 1$, the normalization specified in the conjecture is

$$\tilde{T} = aN.$$

Choose

$$F = bN.$$

Then F has rank one, $F^2 = 0$, and $\|F\| = b$, so it is one of the perturbations quantified over by the conjecture.

Let $E = \text{span}\{e_1, e_2\}$. Relative to the ordered basis (e_1, e_2) , the adjoint-side perturbation restricts to

$$(\tilde{T}^* + \alpha F)|_E = aN^* + \alpha bN = \begin{pmatrix} 0 & \alpha b \\ a & 0 \end{pmatrix}.$$

Its square is

$$\begin{pmatrix} 0 & \alpha b \\ a & 0 \end{pmatrix}^2 = \alpha ab I_E.$$

Consequently, for every $\alpha \neq 0$ the restriction has the two nonzero eigenvalues satisfying $\lambda^2 = \alpha ab$. Hence $\tilde{T}^* + \alpha F$ is not quasinilpotent for every nonzero α . In particular, the premise of Conjecture 2.1 holds for more than two distinct values of α .

On the non-adjoint side, however,

$$\tilde{T} + \alpha F = (a + \alpha b)N, \quad (\tilde{T} + \alpha F)^2 = 0.$$

Therefore

$$\sigma(\tilde{T} + \alpha F) = \{0\} \quad \text{for every } \alpha \in \mathbb{C}.$$

For any admissible ball radius r , its spectral thickening is

$$\sigma(\tilde{T} + \alpha F) + B(0, r) = B(0, r),$$

which is connected. If the open-ball convention $B(0, 0) = \emptyset$ is used at radius zero, the required inclusion of the nonempty pseudospectrum already fails; under the singleton convention the thickening is still connected. Thus condition (ii) of the conjecture fails for every nonzero α , regardless of the choice of the function Φ .

The hypothesis and condition (i) hold for all nonzero α , but no set S with $|S| \geq 2$ can satisfy condition (ii). Since $a, b, t > 0$ were arbitrary, no existential triple works for $T = N$. This disproves the conjecture. $\square$

4 Consequences and limitations

Theorem 1 answers the source conjecture negatively and blocks the paper’s conditional deductions insofar as they rely on Conjecture 2.1 exactly as printed. It does not solve the Hilbert-space invariant subspace problem. It also does not address a modified statement in which the hypothesis and the pseudospectral conclusion use the same adjoint orientation; the present construction is tailored to the printed asymmetry.

The value of t and the detailed definition of the radius function Φ play no role: once the non-adjoint spectrum is $\{0\}$, every metric thickening of it is connected. The example is finite rank but acts on the infinite-dimensional space required by the source, by extending the 2×2 block by zero.

5 Verification and novelty

The proof audit in `VERIFICATION.md` checks the source quantifiers, rank and norm conditions, both perturbation orientations, and the final connectedness obstruction. The script `code/check_matrices.py` checks 36 numerical instances of the two displayed matrix identities; it is only a sanity check, not part of the proof.

The local run indexes were searched for arXiv:2008.03253, the exact title, rank-one quasinilpotent perturbations, and invariant-subspace terms. Bounded web searches on August 11, 2026 used the exact source title together with “counterexample,” the arXiv id together with “Conjecture 2.1,” and the author together with the quasinilpotent invariant-subspace conjecture. They found the current source version and adjacent invariant-subspace papers, but no later answer or this counterexample. The official arXiv record lists v2, revised August 25, 2021, as current. Novelty confidence is positive but conservative pending expert review.

Human review is strongly recommended. The decisive audit is the literal adjoint orientation in Figure 1; after that, the proof is elementary two-dimensional operator algebra.

References

- [1] M. Norman, *A conditional proof of the ISP for quasinilpotent operators*, arXiv:2008.03253v2 [math.FA], 2021.